Requirement Gathering

HDI Prediction System | Requirement Analysis Phase

Purpose

This document provides a clear understanding of the project objectives and requirements. It forms the foundation for designing, developing, testing, and documenting the Human Development Index Prediction System.

Stakeholders

- End Users – students, researchers, and policy enthusiasts who want quick HDI estimates.
- Project Team – responsible for data preparation, model training, and web app development.
- Mentor / Evaluator – reviews project milestones and deliverables.

Functional Requirements

ID	Requirement
FR-1	The system shall accept four numeric inputs: life expectancy at birth, expected years of schooling, mean years of schooli
FR-2	The system shall predict a continuous HDI score in the range [0, 1] using a trained regression model (HDI.pkl).
FR-3	The system shall classify the predicted score into a UNDP development tier: Very High (>= 0.800), High (>= 0.700), Med
FR-4	The system shall expose a web form (index.html) that renders the prediction result on the same page.
FR-5	The system shall expose a JSON API endpoint (POST /predict) for programmatic access.
FR-6	The system shall validate input and return a clear error message when values are missing or non-numeric.

Non-Functional Requirements

ID	Requirement
NFR-1	Usability – a first-time user should be able to obtain a prediction without external instructions.
NFR-2	Performance – a prediction should be returned in well under 1 second on typical hardware, since the model is a
NFR-3	Portability – the application shall run locally via Flask's development server and be deployable via Gunicorn on a
NFR-4	Reliability – invalid input must not crash the server; errors are caught and surfaced to the user.
NFR-5	Maintainability – the trained model and the serving code are decoupled (HDI.pkl vs. app.py) so the model can be

Data Requirements

The project uses the UNDP Human Development Index dataset (HDI.csv), containing 195+ countries with indicators spanning 1990 to 2021, including HDI, life expectancy, expected and mean years of schooling, GNI per capita, and several inequality-adjusted and gender-disaggregated indices. For the prediction model, four core columns are used as features: **Life Expectancy at Birth**, **Expected**

Years of Schooling, **Mean Years of Schooling**, and **Gross National Income Per Capita**, with **Human Development Index** as the target variable.

Constraints

- The model must only use the four indicators available on the input form (no derived or inequality-adjusted features at inference time).
- The application must work without a persistent database; the model is loaded once at server startup from HDI.pkl.